

GROUP THEORY. — Non-residually finite central extensions of arithmetic groups. Note (*) by Mr. Pierre Deligne, Member of the Academy.

[[English translation of the French abstract]]

A variant of known methods for treating the congruence subgroup problem gives results such as the following: for $n \geq 2$, the group $\widetilde{Sp}(2n, \mathbb{Z})$, the inverse image of $Sp(2n, \mathbb{Z})$ in the universal covering of $Sp(2n, \mathbb{R})$ (a central extension of $Sp(2n, \mathbb{Z})$ by $\pi_1 Sp(2n, \mathbb{R}) = \mathbb{Z}$), is not Hausdorff for the topology defined by finite-index subgroups. More precisely, every finite-index subgroup of $\widetilde{Sp}(2n, \mathbb{Z})$ contains $2\pi_1 Sp(2n, \mathbb{R})$.

[[English abstract printed in the 1978 source, retained as source matter]]

A variant of known methods to handle the congruence subgroup problem yields results like the following. Let $\widetilde{Sp}(2n, \mathbb{Z})$ be the inverse image of $Sp(2n, \mathbb{Z})$ in the universal covering of $Sp(2n, \mathbb{R})$; it is a central extension of $Sp(2n, \mathbb{Z})$ by $\pi_1 Sp(2n, \mathbb{R}) \simeq \mathbb{Z}$. If $n \geq 2$, the group $\widetilde{Sp}(2n, \mathbb{Z})$ is not residually finite : any subgroup of finite index contains $2\pi_1 Sp(2n, \mathbb{R})$.

1. S-ARITHMETIC GROUPS. — Let G be an almost simple, simply connected group isotropic over a global field F . To simplify the statements, we shall also suppose that $G(F)$ is perfect (i.e. equal to its commutator subgroup); this is the case except, perhaps, when G is a trialitarian form of D_4 . Let S also be a finite set of places of F containing all the infinite places, and suppose that

$$(i) \sum_{v \in S} \text{rank}(G/F_v) \geq 2.$$

For each place v of F , put $G_v = G(F_v)$. For each finite set T of places, likewise put $G_T = \prod_{v \in T} G_v$ and $G_{T'} = \prod_{v \notin T} G_v$ (restricted product): for $\mathbb{A}$, the ring of adèles of F , one has $G(\mathbb{A}) = G_T \times G_{T'}$.

A subgroup of $G(F)$ is called an S -congruence subgroup if it is the inverse image of a compact open subgroup of $G_{S'}$. It is called S -arithmetic if it is commensurable with an S -congruence subgroup, i.e. if it contains, with finite index, a subgroup of finite index in an S -congruence group. The S -congruence subgroups (respectively, S -arithmetic subgroups) form a neighborhood basis of the identity for a topology (of a topological group) $\mathcal{T}(c)$ [respectively, $\mathcal{T}(a)$] on $G(F)$. For a fixed S -congruence subgroup Γ , one still obtains a neighborhood basis by considering only the S -congruence subgroups normal in Γ (respectively, only the normal finite-index subgroups of Γ). This ensures the existence of the completion $\widehat{G(F)}(c)$ [respectively, $\widehat{G(F)}(a)$] of $G(F)$ for $\mathcal{T}(c)$ [respectively, $\mathcal{T}(a)$]. The topology $\mathcal{T}(c)$ is the topology induced from that of $G_{S'}$. Since $G(F)$ is dense in $G_{S'}$ (the strong approximation theorem), one has $\widehat{G(F)}(c) = G_{S'}$. The completion $\widehat{G(F)}(a)$ of $G(F)$ is an extension of $\widehat{G(F)}(c) = G_{S'}$ by the profinite group $C(S, G) = \lim \text{proj } \Gamma/\Delta$ (the projective limit over pairs $\Delta \subset \Gamma$, with Δ normal of finite index in Γ and Γ an S -congruence group).

For G of rank ≥ 2 , Raghunathan (1) proved that:

$$(ii) \widehat{G(F)}(a) \text{ is a central extension of } \widehat{G(F)}(c) = G_{S'},$$

and he completes the verification that the same conclusion holds under hypothesis (i) alone.

Let $\widetilde{G}_S$ be a topological central extension of G_S by a discrete group A , and let $\widetilde{G(F)}$ be the inverse image of $G(F)$ in $\widetilde{G}_S$. A subgroup of $\widetilde{G(F)}$ will be called an S -congruence subgroup if it is the inverse image of an S -congruence subgroup of $G(F)$, and S -arithmetic if it is commensurable with an S -congruence subgroup. We again denote the corresponding topologies by $\mathcal{T}(c)$ and $\mathcal{T}(a)$, and the separated completions by $\widehat{\widetilde{G(F)}}(c)$, $\widehat{\widetilde{G(F)}}(a)$, for these

topologies. One again has $\widehat{G(F)}(c) = G_{S'}$, while $\widehat{G(F)}(a)$ is an extension of $G_{S'}$ by a profinite group $\widetilde{C}(S, G)$, itself an extension of $C(S, G)$ by

$$\widehat{A} = \lim \text{proj } A/(A \cap \Gamma) \quad (\Gamma \text{ S-arithmetic}).$$

PROPOSITION 1. — If condition (ii) holds, $\widehat{G(F)}(a)$ is a central extension of $\widehat{G(F)}(c) = G_{S'}$.

Recall that if a group $\widetilde{H}$ is a central extension of a group H , the commutator map $\widetilde{H} \times \widetilde{H} \rightarrow \widetilde{H}$ factors through a map, again denoted $(,) : H \times H \rightarrow \widetilde{H}$. In particular, since $\widehat{G(F)}(a)$ is a central extension of $\widehat{G(F)}(c)$ by $\widehat{A}$, the commutator map $\widetilde{C}(S, G) \times \widehat{G(F)}(a) \rightarrow \widehat{G(F)}(a)$ factors through $C(S, G) \times \widehat{G(F)}(a)$. By (ii), it factors through a bimultiplicative map $C(S, G) \times \widehat{G(F)}(a) \rightarrow \widehat{A}$;

since $G(F)$ is perfect and dense in $\widehat{G(F)}(a)$, this map is trivial.

More concretely, the proposition means that for every S-arithmetic subgroup Δ one has:

(1 a) For every conjugate Δ^g of Δ ($g \in \widehat{G(F)}$), there exists an S-congruence group Γ such that $\Gamma \cap \Delta = \Gamma \cap \Delta^g$.

(1 b) $\widehat{G(F)}$ acts trivially on the finite group $C_\Delta = \lim \text{proj } \Gamma/(\Gamma \cap \Delta)$.

These conditions are also equivalent to:

(1 c) For every $g \in G(F)$, there exists an S-congruence group Γ such that $(g, \Gamma) \subset \Delta$.

One has $\widetilde{C}(S, G) = \lim \text{proj } C_\Delta$.

2. CENTRAL EXTENSIONS OF ADELIC GROUPS. — From now on we shall suppose that:

(iii) the group $\widetilde{G}_S$ is perfect.

The most interesting case would be that in which $\widetilde{G}_S$ is the universal central extension of G_S by a discrete group (the “universal covering” of G_S), but we do not wish to assume the existence of such a universal extension.

A topological central extension E of $G(\mathbb{A})$ by a discrete group X is determined by the extensions E_S and $E_{S'}$ of G_S and $G_{S'}$ by X , obtained as the inverse images of G_S and $G_{S'}$ in E : because G_S is perfect, one has

$$E \leftarrow (E_S \times E_{S'}) / \{(x, -x) \mid x \in X\}.$$

If the extension E_S is dominated by $\widetilde{G}_S$, i.e. is obtained from $\widetilde{G}_S$ by pushout along $f : A \rightarrow X$, then f is unique, and the data of E amount to the data of the extension $E_{S'}$ together with f (an equivalence of categories); one has

$$E \leftarrow \widetilde{G}_S \times E_{S'} / \{(a, -f(a)) \mid a \in A\}.$$

Let $\mathcal{L}$ be the category of topological central extensions of $G(\mathbb{A})$ by a finite group, split over $G(F)$, and with E_S dominated by $\widetilde{G}_S$:

$$1 \rightarrow X \rightarrow E \rightarrow G(\mathbb{A}) \rightarrow 1$$

$$\nwarrow_s \uparrow$$

$$G(F)$$

It would amount to the same thing to require “splittable over $G(F)$ ”: since $G(F)$ is perfect, the splitting, if it exists, is unique. The category $\mathcal{L}$ is filtered on the left. For (E, s) in $\mathcal{L}$, the closure in E of $(\text{image of } \widetilde{G}_S) \cdot sG(F)$ maps onto $G(\mathbb{A})$: its image is closed, because X is finite, and it contains $G_S \cdot G(F)$, which is dense (strong approximation). The subcategory L of $\mathcal{L}$ consisting of the pairs (E, s) for which $(\text{image of } \widetilde{G}_S) \cdot sG(F)$ is dense in E is therefore cofinal; since there is at most one arrow between any two objects of L , it reduces to a filtered ordered set.

PROPOSITION 2. — Under hypotheses (ii) and (iii), we construct a natural isomorphism between $\widetilde{C}(S, G)$ and the projective limit over $\mathcal{L}$, or L , $\lim \text{proj } X$. Under this isomorphism, the natural map from A to $\widetilde{C}(S, G)$ is the limit of the maps $-f : A \rightarrow X$, where f is such that E_S is obtained from $\widetilde{G}_S$ by f .

Let E be a topological central extension of $G(\mathbb{A})$ by a discrete group X . Giving a splitting s of E over $G(F)$ amounts to giving an isomorphism between the extension of $G(F)$ by X induced from E_S and the opposite of the extension induced from $E_{S'}$. When E_S is dominated by $\widetilde{G}_S$ and f is as above, this amounts to giving a commutative diagram

$$\begin{array}{ccccccc} 1 & \rightarrow & X & \rightarrow & E_{S'} & \rightarrow & G_{S'} \rightarrow 1 \\ \uparrow & -f & \uparrow & & \uparrow & & \uparrow \\ 1 & \rightarrow & A & \rightarrow & \widetilde{G(F)} & \rightarrow & G(F) \rightarrow 1. \end{array}$$

Hence there is an equivalence between the category $\mathcal{L}$ and the category $\mathcal{L}'$ of commutative diagrams

$$\begin{array}{ccccccc} (1) & 1 & \rightarrow & X & \rightarrow & E_{S'} & \xrightarrow{-q} G_{S'} \rightarrow 1 \\ & \uparrow & & \uparrow & & \uparrow & \\ & \widetilde{G(F)} & \rightarrow & G(F) & & & \end{array}$$

(where $E_{S'}$ is a topological central extension of $G_{S'}$ by a finite group): E is reconstructed as $\widetilde{G}_S \times E_{S'} / \{(a, ta) \mid a \in A\}$, and the splitting over $G(F)$ is given by $s(g) = (\tilde{g}, t\tilde{g})$ modulo the elements (a, ta) , for $\tilde{g} \in \widetilde{G(F)} \subset \widetilde{G}_S$ mapping to g in $G(F)$. The subcategory L of $\mathcal{L}$ corresponds to the subcategory L' of diagrams (1) in which t has dense image.

Let $(E_{S'}, t)$ be in L' , and let $\mathcal{T}$ be the topology on $\widetilde{G(F)}$ induced by that of $E_{S'}$. If K is a compact open subgroup of $E_{S'}$ disjoint from X , the intersections of $\Delta = t^{-1}K$ with the S -congruence subgroups form a neighborhood basis for $\mathcal{T}$. For Γ an S -congruence subgroup defined by a compact open subgroup $K_1 \subset qK$, one furthermore has

$$(2) \quad X \twoheadrightarrow XK/K \leftarrow \sim q^{-1}K_1/(K \cap q^{-1}K_1) \leftarrow \sim \Gamma/(\Delta \cap \Gamma) \leftarrow \sim C_\Delta.$$

Since X is finite, Δ is S -arithmetic. Conversely, Proposition 1 ensures that a neighborhood basis of S -arithmetic groups is obtained in this way: the filtered ordered set L' identifies with the set of germs (for the S -congruence topology) of S -arithmetic groups, and the isomorphism of Proposition 2 is defined as the projective limit of the isomorphisms (2).

3. QUASI-SPLIT GROUPS. — Let μ be the group of roots of unity in F and; for every non-complex place v of F , let μ_v be the group of roots of unity in F_v . For complex v , put $\mu_v = \{1\}$. Deodhar proved that, for G quasi-split, there exists a universal topological central extension $\widetilde{G}_v$ of G_v by a discrete group (the “universal covering”), and that for v non-real its kernel $\pi_1(G_v)$ is canonically a quotient of μ_v . I completed this result by proving that in fact $\pi_1(G_v) = \mu_v$. For real v , one has $\pi_1(G_v) = \mathbb{Z}$ or $\mathbb{Z}/2$. In every case, μ_v occurs as a quotient of $\pi_1(G_v)$.

For almost every v , the extension $\widetilde{G}_v$ splits over the natural maximal compact subgroup of G_v . This makes it possible to define the restricted product of the $\widetilde{G}_v$, an extension of $G(\mathbb{A})$ by the direct sum of the $\pi_1(G_v)$. Push it out by

$$R : \bigoplus_v \pi_1(G_v) \rightarrow \bigoplus_v \mu_v \xrightarrow{-\sigma} \mu : \sigma((x_v)) = \prod_v x_v^{[\mu_v : \mu]}.$$

This gives a central extension of $G(\mathbb{A})$ by μ , and it follows from Deodhar (2) and C. Moore (3) that this is the universal topological central extension of $G(\mathbb{A})$ by a discrete group, split over $G(\mathbb{F})$:

$$(3) \quad 1 \rightarrow \mu \rightarrow \widetilde{G(\mathbb{A})} \rightarrow G(\mathbb{A}) \rightarrow 1$$

$\nwarrow \uparrow$

$$G(\mathbb{F})$$

Take $\widetilde{G_S}$ to be the universal covering of G_S , and let R_S be the restriction of R to $\pi_1 G_S = \prod_{v \in S} \pi_1(G_v)$. Proposition 2 identifies the completion $\widetilde{G(F)(a)}$ of $\widetilde{G(F)}$ with the inverse image $\widetilde{G(\mathbb{A})}_{S'}$ of $G_{S'} = \widetilde{G(F)(c)}$ in (3), and gives an exact commutative diagram

$$(4) \quad 1 \rightarrow \mu \rightarrow \widetilde{G(\mathbb{A})}_{S'} \rightarrow G_{S'} \rightarrow 1$$

$$\uparrow -R_S \uparrow i \uparrow$$

$$\prod_{v \in S} \pi_1(G_v) \rightarrow \widetilde{G(F)(a)} \rightarrow \widetilde{G(F)(a)} \rightarrow 1$$

In particular:

THEOREM. — Let $G/\mathbb{F}$ be quasi-split, and let $\widetilde{G(F)}$ be the inverse image of $G(\mathbb{F})$ in the universal covering of G_S . If $\sum_{v \in S} \text{rank}(G_v) \geq 2$, the intersection of the S-arithmetic subgroups of $\widetilde{G(F)}$ is the kernel of $R_S : \prod_{v \in S} \pi_1(G_v) \rightarrow \mu$.

For $\text{Sp}(2n, \mathbb{Q})$ and $S = \infty$, this is the result announced in the introduction. For $\text{SL}(n, K)$, with K real quadratic and $S = \infty$, the resulting statement generalizes that of J. Millson (4).

PROBLEM. — Does every discrete finite-covolume subgroup of the universal covering of $\text{Sp}(2n, \mathbb{R})$ ($n \geq 2$) contain $2\pi_1 \text{Sp}(2n, \mathbb{R})$? (5).

4. THE SYSTEM OF FACTORS OF THE θ FUNCTIONS. — When condition (i) is not satisfied, Proposition 1 is no longer available; it nevertheless remains possible to study S-arithmetic subgroups satisfying a condition of type (1 a).

Let S therefore be a nonempty finite set of places containing all the infinite places, and let $T \supset S$ be such that (G, T) satisfies condition (ii). The interesting case is that in which S consists of a single place at which G has rank 1. Let $\widetilde{G_S}$ again be a topological central extension of G_S by a discrete group, and let $\widetilde{G(F)}$ be the inverse image of $G(\mathbb{F})$ in $\widetilde{G_S}$. For an S-arithmetic subgroup Δ of $\widetilde{G(F)}$, consider the condition:

($\star$) there exists a T-arithmetic subgroup U of $\widetilde{G(F)}$ such that, for every conjugate Δ^g of Δ ($g \in U$), there exists an S-congruence group Γ for which $\Gamma \cap \Delta = \Gamma \cap \Delta^g$ (U-invariance of the germ of Δ).

PROPOSITION 3. — The completion $\widetilde{G(F)(a^*)}$ of $\widetilde{G(F)}$ for the topology of S-arithmetic subgroups satisfying ($\star$) is a central extension of $\widetilde{G(F)(c)} = G_{S'}$.

More concretely, we must verify that if (U, Δ) satisfies ($\star$), then Δ satisfies (1 c). To prove this, we may first shrink U and Δ . In particular, we may suppose that $\Delta \subset U$ (replacing Δ by $\Delta \cap U$). The group U then acts by conjugation on the

finite group $C_\Delta^U = \lim \text{proj } (\Gamma \cap U)/(\Gamma \cap \Delta)$ (Γ S-congruence); replacing U by the kernel of this action and Δ by some $\Delta \cap \Gamma$, we may further suppose that U acts trivially on C_Δ^U .

Since U is T-arithmetic, its completion $U(c) \subset G_{S'}$ has finite index in a subgroup $G_{T-S} \times L$, with $L \subset G_{T'}$ compact open; since G_{T-S} has no open subgroup of finite index, $U(c)$ itself has the form $G_{T-S} \times L$. Condition $(\star)$ ensures that the subgroups $\Delta \cap \Gamma$ (Γ S-congruence) form a neighborhood basis of the identity for a group topology on U . The corresponding completion $U(\Delta)$ is an extension of $U(c)$ by C_Δ^U —and hence a central extension of $G_{T-S} \times L$. Replace L by an open subgroup L_1 over which this extension is trivial, and replace U and Δ by the inverse images of L_1 . Since G_{T-S} is perfect, a central extension of $G_{T-S} \times L$ is determined by its restrictions to the two factors; this reduces us to the case in which the central extension $U(\Delta)$ of $G_{T-S} \times L$ is the inverse image of a central extension $\widetilde{G}_{T-S}$ of G_{T-S} by C_Δ^U . By construction, U lifts to $\widetilde{G}_{T-S}$, and Δ is the inverse image of a compact open subgroup K of $\widetilde{G}_{T-S} \times L$:

$$(5) \quad K \hookrightarrow \widetilde{G}_{T-S} \times L$$

$\uparrow \uparrow$

$$\Delta \longrightarrow U$$

Let $\widetilde{G(F)}_T$ be the central extension of $G(F)$ obtained as the inverse image of $G(F)$ in $\widetilde{G}_T = \widetilde{G}_S \times \widetilde{G}_{T-S}$, and let $\widetilde{U}$ be the lifting (5) of U in $\widetilde{G(F)}_T$. For $g \in G(F)$, Proposition 1 gives $(g, \Gamma_T) \subset \widetilde{U}$ for a T-congruence group Γ_T defined by a sufficiently small $L' \subset L$. If the S-congruence subgroup Γ defined by $K' \subset G_{T-S} \times L'$, whose inverse image in $\widetilde{G}_{T-S} \times L'$ is K'' , is small enough that $(g, K') \subset K$, then $(g, \Gamma) \subset \Delta$, which verifies (1 c).

The kernel $\widetilde{C}^*(S, G) = \text{Ker}(\widetilde{G(F)}(a^*) \rightarrow \widetilde{G(F)}(c))$ is still described by Proposition 2 (with the same proof), and the calculation can be carried through in the quasi-split case (cf. §3). Here is an application, with $G = \text{SL}(2, \mathbb{Q})$, $S = \{\infty\}$, and $T = \{\infty, p\}$. Let $\widetilde{SL(2, \mathbb{R})}$ be the double covering of $\text{SL}(2, \mathbb{R})$. The factor systems of the θ functions, restricted to arbitrarily small congruence subgroups of $\text{SL}(2, \mathbb{Z})$, may be interpreted as defining a germ τ (for the congruence-subgroup topology) of liftings of $\text{SL}(2, \mathbb{Z})$ into $\widetilde{SL(2, \mathbb{R})}$. This germ is invariant under $\text{GL}(2, \mathbb{Q})$. If a modular form of half-integral weight $k/2$ is defined to be a function on the Poincaré half-plane which, under the action of a congruence subgroup of $\text{SL}(2, \mathbb{Z})$, transforms like the θ function of a quadratic form in k variables, this amounts to saying that if

(a b)

(c d) $\in \text{GL}(2, \mathbb{Q})$

has positive determinant and f is modular of weight $k/2$, then $f(-z^{-1})$ and $(cz + d)^{k/2} f[(az + b)/(cz + d)]$ are likewise modular. Conversely:

PROPOSITION 4. — For every prime p , the germ τ is the unique germ of liftings of $\text{SL}(2, \mathbb{Z})$ into $\widetilde{SL(2, \mathbb{R})}$ that is invariant under a finite-index subgroup of $\text{SL}(2, \mathbb{Z}[1/p])$.

The arguments leading to the theorem of §3 show that every arithmetic subgroup of $\text{SL}(2, \mathbb{Z})$ whose germ is invariant under a finite-index subgroup of $\text{SL}(2, \mathbb{Z}[1/p])$ is a congruence subgroup.

Let $\Gamma \subset \text{SL}(2, \mathbb{Z})$ be a congruence subgroup, and let $\tau_1 = \tau_2$ be two liftings of Γ into $\widetilde{SL(2, \mathbb{R})}$.

Let Δ be the subgroup of Γ on which $\tau_1 = \tau_2$. If the germs of τ_1 and τ_2 are distinct and invariant under a $\{p\}$ -arithmetic group, then Δ satisfies $(\star)$ and is not a congruence subgroup.

(*) Session of July 3, 1978.

(1) M. S. Raghunathan, Publ. Math. I.H.E.S., 46, 1976, pp. 107-162.

(2) V. V. Deodhar, On central extensions of rational points of algebraic groups (forthcoming).

(3) C. C. Moore, Publ. Math. I.H.E.S., 35, 1968, pp. 5-70.

(4) J. Millson, Real Vector Bundles with Discrete Structure Groups (forthcoming).

(5) I learn from Mostow that such a subgroup Γ contains, in any case, a finite-index subgroup of $\pi_1 \operatorname{Sp}(2n, \mathbb{R})$: one verifies that the projection of Γ into $\operatorname{Sp}(2n, \mathbb{R})$ is discrete, since otherwise the Lie algebra of its closure would be nontrivial and central.

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